Chapter 12

Wood as Cultural Heritage Material and its Deterioration by Biotic and Abiotic Agents

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WOODEN CULTURAL HERITAGES AND THEIR PROPERTY DIAGNOSIS

Wooden Cultural Heritages

Wood has been widely used in structural applications as well as for decorative, ritual, and religious purposes. Intimate human links to wood over thousands of years are embedded in wooden cultural heritage (WCH) representing human life and values. Stone, bronze, and iron have their own epoch of utilization but

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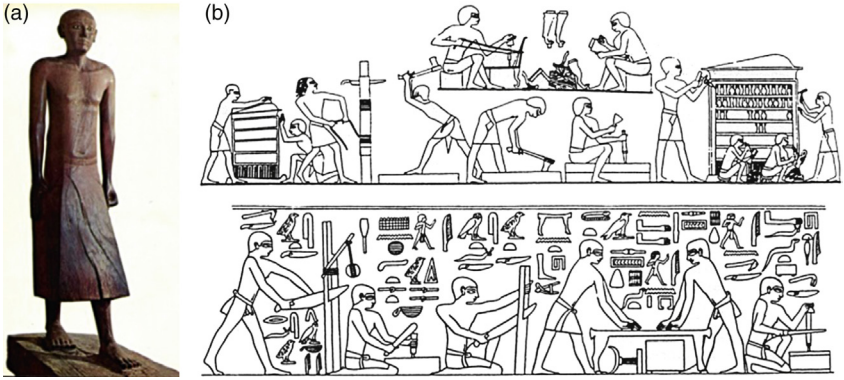


FIGURE 12.1 (a) Statue of the Chancellor Nakt, Egyptian middle kingdom at the time of 13th dynasty (Louvre Museum) and (b) Relief on the Tombs of Tebe (1400 BC) and Saqquara (2500 BC) showing the utilization of wood by the Egyptians. *(Wooden statue from "Masterpieces of the Louvre and the Jeu de Paume", Casa Editrice Bonechi, Florence, 1974 and Relief on the tomb from Kuenhn H, second. ed. "Erhalting und Pflege von Kunstwerken und Antiquitaeten". Keyer, Munich, 1981.)*

wood has been continuously used throughout history. There is no indication of the direct use of wood by the earliest hominids, although Neanderthals exhibited many complex behaviors such as funerary practices (d'Errico et al., 2012). Cultural uses of wood appear to be related to the activity of humans rather than hominids (Bamford, 2010) (Fig. 12.1).

Any wooden artifact that provides us information about human life and culture, and that is considered worthy of preservation for the future can be defined as wooden cultural heritage (Rowell and Barbour, 1990). Vocational, technical artifacts, creative activities, and cannons of craftsmanship are embedded in WCH, reflecting past human culture, ideals, and symbols. Thus, the technological and sociological aspects of human activities can be understood from excavating WCH objects or remains. Consequently, a holistic approach comprising of natural sciences, humanities, and the arts is needed for a meaningful and informative study of WCH.

WCH as tangible objects can be generally classified into three groups: (1) moveable, (2) immoveable, and (3) underwater (according to UNESCO). Moveable WCH include musical instruments, painted panels, furniture, Italian intarsia, sculptures, iconic altar, Tripitaka-plates, coffins, and masks. In general, moveable WCH are displayed and/or kept indoors with air-conditioning. Immoveable WCH include Buddhist temples, churches, chapels, royal palaces, wooden pagodas, watermills, windmills, and wooden bridges of historic value. Most immoveable WCH are situated under outdoor conditions without air-conditioning and are exposed to fluctuations in temperature and humidity (Fig. 12.2). In contrast, underwater WCH include shipwrecks, foundation piles,

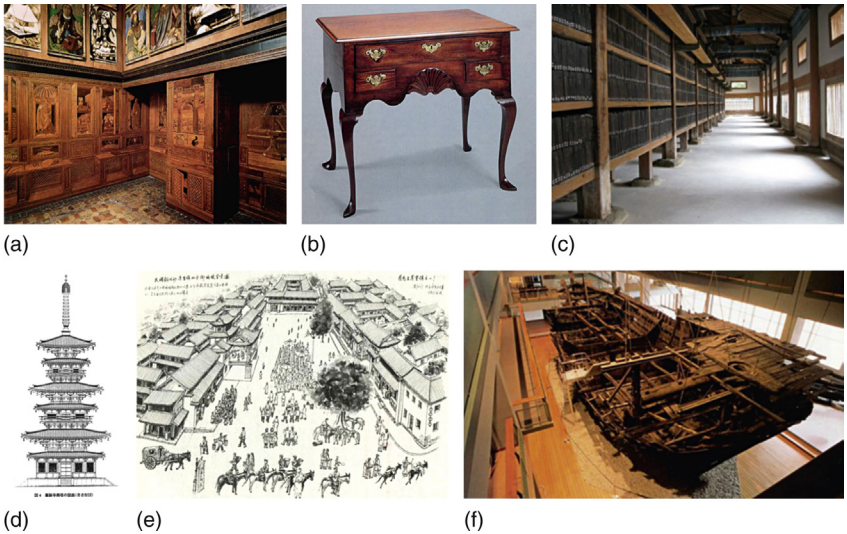


FIGURE 12.2 Typical wooden cultural heritages as movable, immoveable, and underwater WCH. (a) Studiolo in Florence, (b) dressing table in 1740–1750 in the United States, (c) Tripitaka in Korea as moveable WCH, (d) wooden pagoda in Japan, (e) wooden house village in China as immoveable WCH, (f) sunken ship of Kogge in Germany as underwater WCH. *(Studiolo in Florence and dressing table from "The Metropolitan Museum of Art Bullentin," Fall, 1995; Tripitaka in Korea from Kim, Y.S. et al. (Eds.) "Wood Preservation Science," Chonnam Nat'l Univ Press, Korea, 2004; wood pagoda from Itoh, T. (Ed.), "Wood Culture and Sciene," Kaiseisha Press, Japan, 2008; wooden house village from Liu, P. and Liu, Y. "Yunnan Ancient Architecture," China 2012; and Kogge ship from "Die Kogge," Convent, Hamburg 2003.)*

wooden cargo or contents with a cultural, historical, or archaeological character, which were partially or totally underwater periodically or continuously for at least 100 years.

Property Diagnosis of WCH

Virtually all WCH are in a constant state of chemical transformation and can deteriorate over time. Sometimes deterioration of WCH is relatively slow, either because wood species are quite durable or exposure conditions have been mild and nonaggressive (Koestler et al., 2010). However, many WCH are disintegrating and disappearing at a distressing rate simply from neglect and unintended abuse. The extent of deterioration of WCH varies from near normal appearance to disintegrated and extensively changed wood, depending on the agents of deterioration, wood species, and environmental conditions. Various abiotic and biotic agents can deteriorate WCH (Table 12.1). Exact diagnostic data on the existing state of recovered WCH are crucial not only for their conservation but also for understanding the degradation processes.

TABLE 12.1 Deterioration of WCH by Biotic and Abiotic Agents

Types of WCH	Biotic agents	Abiotic agents
Dry-type WCH	Fungi Insects Termites	Moisture fluctuation Weathering
Wet-type WCH	Fungi Bacteria Marine borers	Hydrolysis

Application of Wood Science in WCH Studies

Modern wood science knowledge can contribute to the diagnosis and conservation of WCH by understanding their anatomical, biological, physical, and chemical characteristics. Identification of wood species, strength analysis, monitoring physical and dimensional changes, and evaluation of the degree of deterioration by microorganisms and insects are the essential tasks for proper evaluation and diagnosis of WCH by wood scientists (Itoh and Yamada, 2012). Furthermore, knowledge of wood science can aid in the implementation of nondestructive methods and guidelines for the conservation of WCH, although the methodologies and techniques employed by wood scientists are not always welcomed by curators, museum authorities, and conservators, who have a different mindset to understanding and conserving WCH. They are concerned with the loss of the originality and integrity of WCH during analytical investigation because the diagnostic methods applied by wood scientists are generally of an invasive or destructive nature. A rational integration of science and art should be developed for anthropological interpretation without damaging the historical value and beauty of WCH.

Identification of wood species is a compulsory step prior to any investigation of WCH because identification of wood species provides information on the human control over the wood and also on the floral ecosystem (Itoh, 2008). For example, identification of wood species by our ancestors contributed to the knowledge of wood durability. The ancient Chinese used durable woods for coffins of their emperors and noble classes (Wang, 2010). Proietti et al. (2011) found that the main wood species of Egyptian sarcophagus was yew (*Taxus* spp.), which was not native to Egypt. The wood was probably imported from Turkey or Northwest Africa because of its high durability, which guaranteed a better preservation and thus improved the mummy conservation of the sarcophagus. Identification of wood species is also essential for the conservation treatment of WCH in order to distinguish the original work from later restorations. The palette of the fifteenth century Florentine intarsia cutter consisted of local woods such as walnut (*Juglans* sp.), pear (*Pyrus* sp.), cherry (*Prunus* sp.), maple (*Acer* sp.), and oak (*Quercus* sp.). However, Wilmering (1996) found that the nineteenth century restorations were

TABLE 12.2 Currently Available Nondestructive Methods

Mechanical	Stiffness, bending, tension, compression, probes	
Sonic/ultrasonic	Wave propagation, acoustic emission	
Electrical	Electrical resistance, dielectric and piezoelectric properties	
Magnetic	Nuclear magnetic resonance	
Electromagnetic and nuclear radiation	Reflection	Visual assessment, IR/NIR spectroscope
	Diffraction	X-ray small-angle scattering
	Transmission	imaging and tomography with X-ray or neutrons; X-ray densitometry
	Emission	IR-thermography
Chemical	Composition	

Mannes (2009).

done mostly in rosewood (*Dalbergia* sp.), maple, artificially stained oak, sycamore (*Platanus* sp.), and fir (*Abies* sp.). Direct visual evaluation of wood species in WCH based on the characteristic macroscopic features is not always possible. Visual identification of WCH is not conclusive due to its biological degradation or an opaque finishing or painting. Although the microscopic approach is more reliable than visual examination, sampling of wood from the WCH for identification is not always permitted. In this regard, trained specialists as well as noninvasive methods should be developed to overcome these difficulties. Recently, the use of two-photon excited fluorescence (2PEF) microscopy produced highly contrasted 3D images at micrometer scale without any preparation or sampling of the artifacts, making *in situ* observations of historical violins possible without any damage to the instrument (Latour et al., 2012). X-ray microtomography was also applied for noninvasive identification of ancient wood objects (Steppe et al., 2004; Mizuno et al., 2010; Maierhofer et al., 2010).

Because most of WCH are rare or irreplaceable objects, the use of noninvasive or nondestructive testing (NDT) is more favored than destructive methods. NDT methods in general do not affect the integrity of the examined objects. However, each NDT method has its own specific limitations, such as low contrast, resolution, sensitivity for hydrogen (water), and accessible sample size (from millimeter to decimeter range). Usually a combination of several different examination or analytical procedures is involved for the study of a given WCH. NDT methods applied in the identification of the current state of WCH are summarized in Table 12.2 (Mannes, 2009).

PHYSICAL AND CHEMICAL CHARACTERISTICS OF WCH

Physical Characteristics of WCH

Largely, the physical and mechanical properties of moveable and immoveable WCH are likely to be the same as for recent wood because the dry conditions of WCH inhibit the activity of most agents of deterioration. Inspections have shown that some old wooden objects, such as Buddhist monasteries and temples, are similar to recent wood in density and mechanical characteristics. In some cases, immovable WCH showed higher density and mechanical strength than recent wood. Kohara (1958) found that the bending strength of 1800-year-old Japanese cypress (softwood) increased for the first 200 years and then gradually decreased to about the same strength as for recent wood, whereas *Zelkoba* spp. (hardwood) showed a gradual decrease in mechanical strength. Changes in the stiffness of softwoods and hardwoods over time can be explained by changes in the crystallinity of cellulose (Nakao et al., 1989) (Fig. 12.3). Recently, Yokoyama et al. (2009) found that aged softwood was brittled especially in the radial direction, where the linear elastic limit was not reached.

Mechanical properties of WCH are affected greatly by moisture content and temperature. In particular, damages resulting from moisture are costly and hazardous. Structural failure due to decay, physical deterioration of the building, loss of thermal insulation, warping, corrosion of metal components, blistering and peeling of protective coatings and finishes, growth of molds, and discoloration are the typical moisture-related systems (Fazio et al., 2010; Irbe et al., 2012; Ortiz et al., 2014; Strzelczyk, 2004). Immoveable WCH, such as

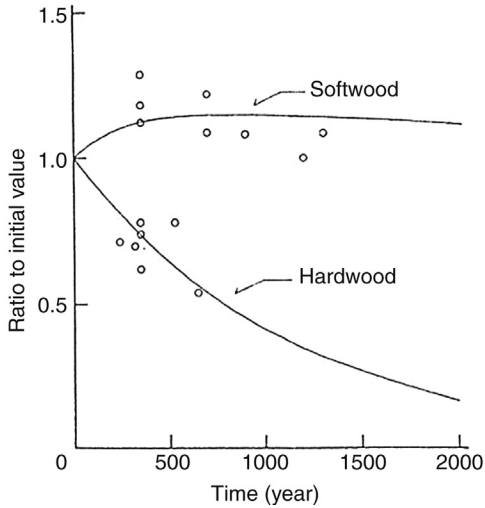


FIGURE 12.3 Changes in the degree of crystallinity and wood stiffness over time. *(Solid lines show degree of crystallinity and open circles denote values of the MOE (Nakao et al., 1989).)*

uninsulated outbuildings, can be severely affected even from a brief exposure to moisture, leading to cracks in musical instruments and veneer lifting. Moveable WCH, such as furniture and musical instruments, in a steam-heated environment (due to low relative humidity) will shrink and crack unless humidification and a long period of acclimatization are provided.

The physical properties and mechanical strength of underwater WCH are inferior to those of recent wood. However, the values of strength decreases are not directly proportional to the losses of woody mass because the deterioration of underwater WCH is not uniform and degradation is usually confined to the surface, particularly in large timbers. Furthermore, individual samples in the underwater WCH usually show great variations in wood properties depending upon the surrounding environmental and burial conditions. Thus, the age of underwater WCH is not a reliable indicator for estimating the degree of their deterioration (Schniwind, 1990). In many cases, the maximum moisture content of waterlogged archaeological wood has been used as a function of strength decreases and dimensional changes.

In general, tensile and bending strength are likely to be much more affected than compression strength parallel to the grain. Dimensional changes of archaeological woods are of great importance because significant shrinkage can occur during drying. The hygroscopicity of underwater WCH sometimes increases drastically. Equilibrium moisture content of the waterlogged ancient wood has been estimated to be more than 50%. Reduced cellulose crystallinity is also responsible for an increase in hygroscopicity and dimensional changes in underwater WCH (Schniwind, 1990) (Table 12.3). However, the degree of dimensional changes depends on the extent of deterioration of underwater WCH. Safe drying of underwater-waterlogged woods is one of the most challenging tasks in the conservation of archaeological woods.

Chemical Characteristics of WCH

It has been reported that usually chemical properties remain unchanged in the moveable and immovable WCH (Takei et al., 1997). In contrast, in underwater

TABLE 12.3 Density and Sorption of Buried Oak Wood

Age years	Residual density (%)	EMC in old wood (%)	EMC in recent wood (%)
570	94	52.0	32.0
1000	75	36.4	30.8
2500	42	60.0	30.0
4700	103	32.0	30.8

Data summarized from Schniwind (1990).

WCH, significant changes in their physical and chemical properties can occur. Chemical characteristics of WCH have been carried out mostly in the underwater WCH, such as waterlogged archaeological woods (Gelbrich et al., 2008; Passialis, 1997). Chemical data on the moveable and immovable WCH are rare because such objects are basically immune to microbial degradation due to their low moisture content.

Major chemical changes in waterlogged archelogical woods (WAW) under long-term waterlogging have been determined by pyrolytic-GC/MS (Tamburini et al., 2014), Fourier transform attenuated total reflectance (FT-MIR-ATR), FT-IR, FT-ATR, FT-NIR (Tsuchikawa et al., 2005; Pedersen et al., 2014), and solid-state nuclear magnetic resonance (NMR) by spectral comparison (Viel et al., 2004). Rapid and accurate estimation of the extent of degradation is important for optimizing restoration and conservation. Wet chemical analyses allow detailed interpretation of the degradation processes due to accuracy and high resolution. However, because of its destructive nature and long examination period, wet analysis is not ideally suited for examining archaeological wooden objects. Recent developments in nondestructive measurements, combining the field of optics (X-ray computed microtomography) and electronics, complement traditional wet chemical methods due to their simplicity of measurement, minimal sample preparation, possibility of repetitions, and highly enhanced reliability and accuracy of the data.

Chemical changes in underwater WCH occurred mainly in structural polysaccharides (Gelbrich et al., 2008; Giachi et al., 2003). Due to the preferential degradation of polysaccharides, the ratio of holocelluose to lignin in the underwater WCH was 0.3∼1:1, and in recent wood 3:1. Not only amorphous but also crystalline cellulose was decomposed during long-term waterlogging situation. Degradation of crystalline cellulose in underwater WCH can be confirmed by X-ray diffraction as well as polarized microscopy, the latter showing loss of the birefringence. Low molecular weight substances were also greatly affected during long-term underwater submersion of WCH. Hot water and 1% alkaline extractives showed a significant increase due to degradation of a large part of low-molecular hydrocarbons derived from hemicelluloses. In contrast, the amount of organic solvent soluble materials decreased. Chemical data of waterlogged bamboo exhibited the same trends as in the waterlogged wood (Table 12.4).

The ash content of the underwater wood and bamboo was greater than that of the recent wood samples partly because of migration of minerals from the sediments into the porous sites in the WAW and partly because of microbial activities (Fig. 12.4). Sulfur accumulation in WAW was induced by bacteria in a seawater environment. Fors et al. (2012) found the presence of sulfate-reducing bacteria (anaerobic) in diverse environments, which reduce sulfates to sulfides and utilize aromatic compounds including lignin and tannins. Some bacteria-degrading WAW in seawater promote the accumulation of sulfur as thiols in the lignin and also in iron sulfides, depending on the conditions.

TABLE 12.4 Chemical Characteristics of Recent and Waterlogged Archeological Wood of *Pinus* spp. and Bamboo

	Holocel-lulose	Lignin	Extractives			Ash
			Hot water	1% alkaline	Alcohol benzene	
Recent pine	73.2	26.85	2.8	12.67	3.0	0.3
Waterlogged *Pinus* spp.	29.47	64.18	7.8	18.73	0.7	7.6
Recent bamboo	75.2	24.3	6.6	25.1	4.4	1.3
Waterlogged bamboo	43.3	62.1	9.8	83.5	2.3	5.4

Kim (1990); Cha et al. (2014).

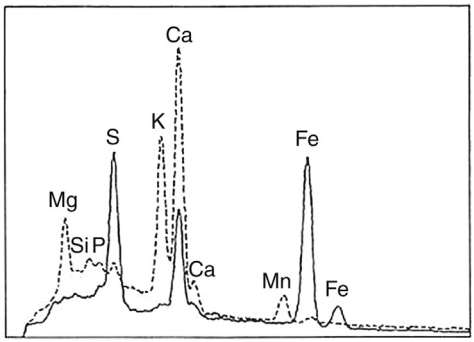


FIGURE 12.4 Diversity in ash present in the waterlogged archeological wood (solid line) in comparison with the recent wood of *Pinusmassonia* (dotted line) (Kim, 1990).

Spectral and signal comparison in FT-IR and NMR indicated a significant decrease in polysaccharides and an increase in lignin in the underwater WCH. FT-IR analyses showed that band intensities assigned to xylan (1730 and 1320 cm^{-1}) and cellulose (1160, 1110, and 1060 cm^{-1}) decreased, whereas band intensities assigned to lignins (1220–1230, 1270 cm^{-1}) significantly increased (Fig. 12.5). Among the polysaccharides, the band assigned to hemicelluloses showed a greater decrease in comparison with cellulose, which suggests preferential degradation of hemicelluloses in the underwater WCH. In contrast, 10,000-year-old spruce lignin revealed no significant structural differences in comparison with the recent spruce wood by quantitative NMR analysis (Christiernin et al., 2009). The data from FT-IR and pyrolysis MS were consistent with results obtained from solid-state NMR (Bardet et al., 2009; Cha et al., 2014; Li et al., 2014).

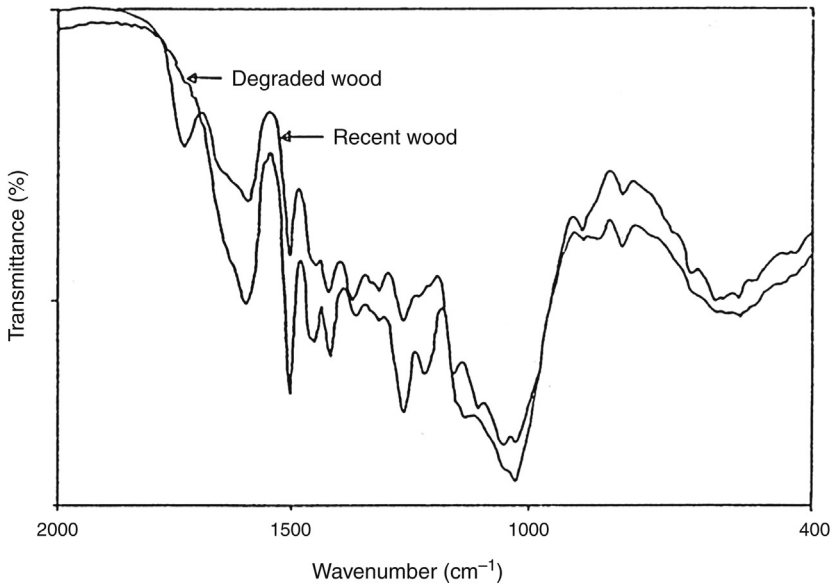


FIGURE 12.5 FT-IR spectra of recent and heavily degraded wood of *Pinus massoniana* (Kim, 1989).

Solid-state NMR analyses of waterlogged bamboo exhibited the same trends as for waterlogged archaeological woods. NMR spectra in the underwater bamboo showed the following: (1) complete disappearance of the signal at 21 ppm assigned to CH_3 of hemicelluloses; (2) a decrease in intensities and/or loss of signals at 64, 74, and 106 ppm, assigned to cellulose; and (3) stronger peak intensities at 105–160 ppm, assigned specifically to the aromatic compounds – the signals assigned to guaiacyl (G) lignin (84 and 115 ppm), syringyl (S) lignin (134, 148, and 152 ppm), and lignin methoxy carbon (56 ppm) showed significant increase. Significant increases in almost all the signals assigned to lignin polymer resulted from the suppression by cellulose and hemicelluloses, which were exposed in the WAW after degradation of structural carbohydrates against microbial attack under waterlogged conditions (Fig. 12.6).

All analyses showed that lignin in WAW is relatively more resistant than structural polysaccharides against microbial attack under waterlogged situations although depolymerization of lignin by cleavage of ether bonds (Salanti et al., 2010; Colombini et al., 2009; Pedersen et al., 2014) and demethoxylation of guaiacyl units was found (van Bergen et al., 2000). Dedic et al. (2014) found that the condition of lignin in the Vasa wood is similar to fresh oak and that potentially harmful tannins are not present in high amounts, suggesting that oxidative degradation mechanisms are not the primary route for cellulose depolymerization in the waterlogged archaeological wooden ships.

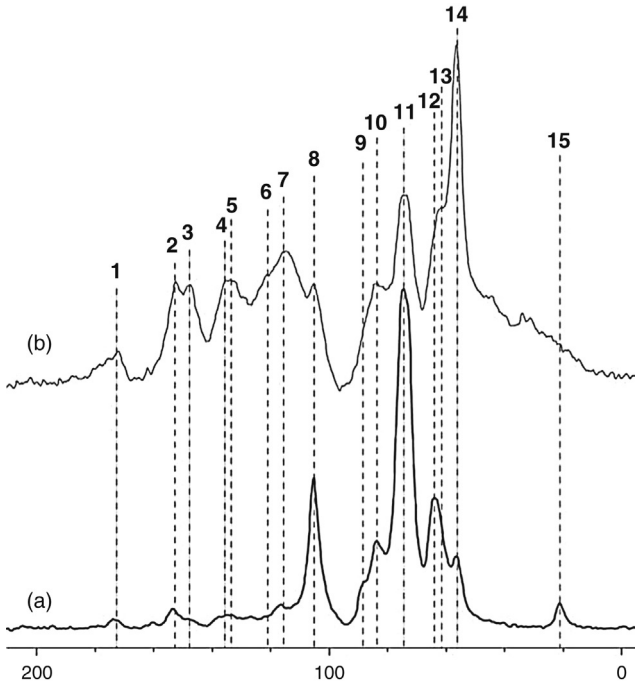


FIGURE 12.6 Solid-state NMR of waterlogged ancient bamboo (b) and recent bamboo (a) (Cha et al., 2014). Resonance assignment: 1, carbohydrate; 2–5, syringyl and guaiacyl lignin; 6 and 7, guaiacyl lignin; 8 and 9, cellulose; 10 and 11, lignins; 12L, carbohydrates; 13, lignins; 14, methoxyl in lignin; 15L, carbohydrates.

ABIOTIC AGENTS IN THE DETERIORATION OF MOVEABLE AND IMMOVEABLE WCH

When moveable and immoveable WCHs are exposed to natural weather conditions, the surface of dry-type WCH becomes rough, the grain rises, and the wood develops checking, warping, and discoloration. Such deterioration results from the process called weathering, which is mainly confined to the surface. Three distinct alterations in the dry-type WCH by weathering are the following: (1) loss of original color, (2) formation of cracks, and (3) erosion of wood mass (Fig. 12.7). Most alterations in the dry-type WCH occur gradually on the surface; the roughened surface ultimately loses its surface coherence and corrugated appearance of the surface is caused by more rapid erosion of earlywood than latewood. The combined effect of light, precipitation, heat, sand, wind, pollutants, and other factors bring about the degradation of exposed WCH to natural weather conditions (Feist, 1990). Biotic agents such as wood-decay fungi and molds are not considered to be primary agents of weathering in dry-type WCH.

FIGURE 12.7 Wood plate in a Buddhist temple exposed to natural weathering in Japan. *(Unpublished file from Kim, Y.S.)*

Discoloration

WCH that are located outdoors in historic buildings and that do not have a surface coating or treatment, undergo rapid changes in color and a loss in their integrity. The surface color of WCH under outdoor conditions changes to brownish color due to oxidation and degradation of lignin. Later wood surface becomes covered with a spongy grayish layer due to the removal of lignin degradation products by rainfall. In contrast, the dry-type WCH stored under indoor conditions exhibit fading of colors by light; uncoated wood turning yellow or red is an indication of fading. The light-colored spruce of a soundboard will darken and the red mahogany of a piano lid will bleach when exposed to light. Discoloration is cumulative, irreversible, and insidious. Discoloration of the unprotected surface can result even from exposure to room light. Lee et al. (2009) found that about 70% of the original color of Japanese cedar (*Cryptomeria japonica*) wooden box was lost in the first year of indoor exposure. However, significantly greater color changes were not found after 5 years. Discoloration is accelerated by moisture content in the air, repetition of drying and wetting, and of hot and cold periods (Matsuo et al., 2008).

Cracks

Dry-type WCH exposed to indoor conditions can exhibit checks or cracks throughout the surface (Fig. 12.8). Cracking and distortion in the WCH derive from the moisture gradient along the wooden objects, producing a differential shrinkage perpendicular to the wood grain. Furthermore, repeated drying and wetting can lead to surface checks and cracks. Warping, tearing, and cracks on the surface of dry-type WCH can occur under very dry conditions. Paint layers, protective varnishes, and lacquers have often been present on dry-type WCH, such as old wooden musical instruments – pianos, harpsichords, and violins.

(a) (b) (c) (d)

FIGURE 12.8 Cracks in various WCH. (a) Buddhist statue (Japan), (b) Mona Lisa panel (Louvre, France), (c) wooden plate (Korea), (d) waterlogged wooden pile (Korea). *(Buddhist statue from "Tokyo Nat'l Museum Bulletin"; Mona Lisa Panel from Uzielli, L., Proc. "The Wood Culture and Science Kyoto, Kyoto 2011"; wooden plate and waterlogged wooden pile from Kim, Y.S.s, unpublished files.)*

Wooden boxes and panels were fastened sometimes by metals. The dissimilar substances expand and shrink disparately with fluctuations in relative humidity and temperature, leading wood and coating materials to swell and shrink differently, resulting in cracks in the wooden part and in the flake-off of paints. Under low humidity and dry conditions, the layers of paints tend to crack more severely. The "Mona Lisa" painted about 500 years ago on one piece of a board of poplar (*Populus alba* L. 790 × 530 mm × 13 mm thick), showed an 11-cm-long crack running through the whole thickness (Fig. 12.8b) (Uzielli, 2011). Even the coating layers, surface treatments, and the original adhesives binding the decorative materials to wooden objects of WCH also deteriorate over time with environmental changes. Such alterations are difficult to repair without changing the visual appearance of the artifact.

Erosion of Wood Mass

It has been observed that erosion in WCH exposed to outdoor conditions occurs mainly in earlywood and that latewood is relatively resistant against weathering. Ultraviolet (UV) radiation in sunlight plays an important role in the disintegration of wood cell wall during the weathering process. Photodegradation in WCH is more or less localized on the surface because of shallow penetration of UV rays into wood. Kim et al. (2008) found that cell wall disintegration caused by weathering starts from the middle lamella, followed by the degradation of the S3 and S1 layers. The S2 layer was the most resistant of the secondary cell wall layers. However, even the S2 layer was degraded in later stages of weathering, caused by progressive detachment of microfibrils due to losses in the cell wall matrix materials, mainly the lignin. In the most advanced stages, the S2 layer also became defibrillated and the cell wall tended to disintegrate from peeling off of bundles of microfibrils (Fig. 12.9).

Erosion of wood surfaces during weathering is a slow process; on average the wood loss is 6 mm in a century, although the rate of surface erosion varies

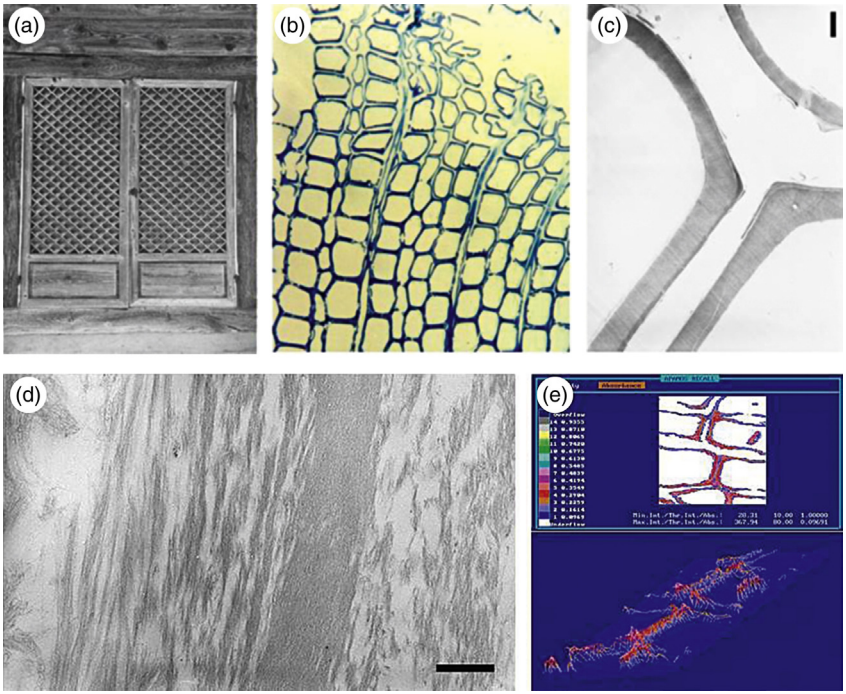


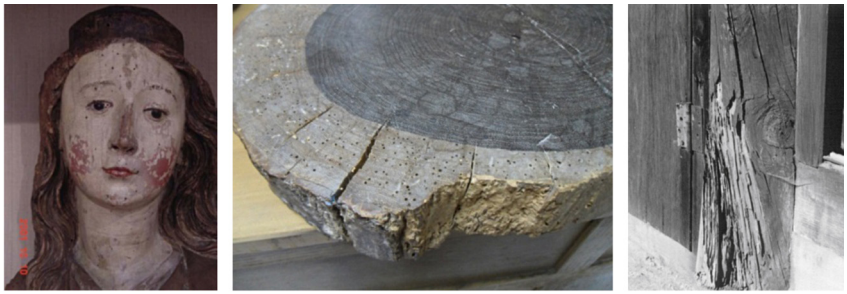
FIGURE 12.9 Disintegration of *Pinus* spp. in the windows of an old Buddhist Temple in Korea exposed to natural weathering (Kim et al., 2008).

between different species (Feist, 1990). In general, hardwoods are more resistant to weathering than softwoods mainly because of their greater density. When dry-type WCH is displayed in a location not directly exposed to sunlight, a much slower erosion of the surface occurs from delignification. Rainfall washes away water-soluble photodegradation products from the wood surfaces, resulting in surface erosion. Mechanical abrasion by wind and sand can aid in the removal of wood mass from degrading wood surfaces.

BIOTIC AGENTS IN THE DETERIORATION OF MOVEABLE AND IMMOVEABLE WCH

Deterioration of Moveable and Immoveable WCH by Insects

Regardless of their moisture content, WCH are prone to insect infestations. Nearly all wood and forest insects are now synanthropic species, which have expanded their niche to include buildings and processed wooden objects (Noldt, 2009; Sutter, 2002; Grosser, 1985). Most insects that attack dry-type WCH belong to the Coleoptera order, the members of which are specialized to attack dried wood with lower moisture content below the fiber saturation point.



(a) (b) (c)

FIGURE 12.10 Damages in indoor WCH by insects. (a) Common furniture beetle in the wooden sculpture in Basel Cathedral, (b) powder-post beetle in the xylarium of Nanjing Forest University, (c) termite attack on the pole of *Pinus* spp. in an old Buddhist temple in Korea. *(Unpublished files from Kim, Y.S.)*

Predominantly, wood-destroying insects in historic buildings in central Europe were found to be the common furniture beetle (*Anobium punctatum*), the deathwatch beetle (*Xestobium nifovillosum*), the anobiid species *Coelostethus pertinnax*, and the European house-borer (*Hylotrupes bajulus*) (Cymorek, 1985; Noldt, 2009; Kisternaya and Kozlov, 2012). The cigarette beetle (*Nicobium hirtum*) was found in a historic building of a Japanese wooden temple (Saito et al., 2008).

Dry-type WCH are also prone to insect infestations even under controlled relative humidity and temperature. Querner et al. (2013) found wood-destroying insects such as *Nicobium castaneum*, the common furniture beetle, *Hexathrum exiguum*, and powder-post beetle (*Lyctus brunneus*) in large museums in central Europe. It is interesting to note that powder-post beetle, which occurs frequently in modern buildings made from imported tropical hardwoods, was found even in the museums in Central Europe. Powder-post beetle is known to attack mainly those species that have large pores, such as oak, ash, walnut, elm, and hickory and bamboo culms containing adequate starch (Fig. 12.10). Softwoods and hardwoods with lower starch content are free from the attack of this beetle. Tropical hardwoods are susceptible to attack by this insect, particularly the light-colored species such as African mahogany, iroko, afzelia, Rhodesian teak, black wattle, ramin, and the many *Shorea* spp., such as seraya, meranti, lauan, and Eucalypts in Australia (Richardson, 1993). Imported household items made from tropical hardwoods, such as furniture, wooden floors, and panels, are often attacked by the powder-post beetle.

In addition to beetles, termite attack was found even in the wooden poles in an old Buddhist temple in South Korea (Fig. 12.10). Termite hazards are confined mainly to tropical and subtropical regions (Nunes, 2010) but are now encountered in New Zealand, Paris in France, and Hamburg in Germany. An increase in transportation and world trade has been responsible for the wide distribution of

TABLE 12.5 Characteristics of Damage by Wood-Boring Insects

	Preferable occurrence in wood species	Length of beetle (mm)	Diameter of exit hole (mm)	Full life cycle (years)
Furniture beetle	Softwood, hardwood, in particular pine, spruce, also poplar, elm, birch	3–6	1.6–2	2–3
Death-watch beetle	Softwood and heartwood primarily in oak wood previously damaged by fungi	6–8	2.5–7	1
Old house horn	Sapwood in softwoods such as pine, spruce and Douglas fir	10–25	10	3–11 or longer
Power post beetle	Sapwood only in hardwood with a certain amount of starch, tropical hardwoods, bamboo	0.8–1.5	0.8–1.6	1 or 2

Data Summarized from Cymorek (1985) and Richardson (1993).

termites. Subterranean termites belonging to genus *Reticulitermes*, *Heterotermes*, and *Coptotermes* (Formosan termites) in the Northern Hemisphere expanded their distribution beyond the latitude 40°N (Richardson, 1993). Subterranean termites attack only the softer earlywood and not the harder latewood, leaving a thin peripheral shell, which poses difficulties in visual inspection. Many foundations in the old Buddhist temples in East Asian countries, constructed with pine (*Pinus* sp.), are prone to termite attacks (Table 12.5).

Deterioration of Moveable and Immoveable WCH by Microorganisms

Dry-type WCHs are considered to be immune to microbial attacks due to their low moisture content. However, some fungal attacks have been observed even in the dry-type WCHs. Some workers (Nilsson and Daniel, 1990; Blanchette, 2010) observed some fungal hyphae in Egyptian coffins, indicating infection and growth of fungi within indoor WCH caused by occasional wetting, leaks, flooding, or other events at some stages. High humidity resulting from flooding and water leakage and poor ventilation in tombs and museums can provide an opportunity for fungal invasion in the dry-type archaeological wooden objects. In particular, molds developed quickly on the wood when wooden furniture, painted panels, and old books were exposed to flooding even though the duration of exposure was short. Fungal colonization over the Tripitaka-Koreana was found due to mistreatment of the plates after ink printing (Park et al., 1996). Dry-type WCHs in

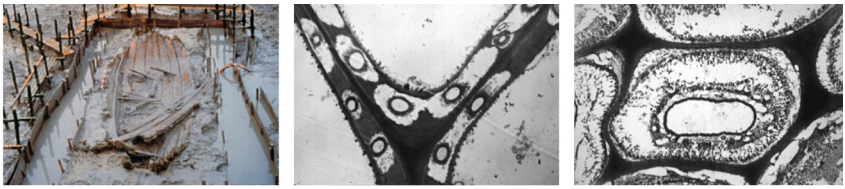
the outdoors have been more prone to a wide range of predators and deleterious factors than those in the indoor because they are frequently exposed to above 70% relative humidity, which can cause molds to readily grow.

When dry-type WCHs are dampened or are resting on a moisture-absorbing surface, such as basement floor, fungal growth and decay can occur. Soft-rot decay has been described frequently in historical and cultural wooden objects from construction timbers in great houses lived in by native Americans, in the ancient Egyptian wooden coffins, in the historic expedition huts in Antarctica (Blanchette, 2000; Blanchette et al., 2004), and in the wooden floor of an old Buddhist temple, which had been treated with fire retardants (Kim et al., 2008). It appears that the moisture-absorbing properties of the fire retardants induced the soft-rot decay in the wooden floors. The occurrence of soft-rot decay in waterlogged archaeological woods has been well documented (Kim and Singh, 2000; Bjoerdal, 2012). Soft-rot fungi belonging to Deuteromycota and Ascomycota are known to be more tolerant to severe growth conditions such as high moisture content, pH, extreme temperature, and high preservative concentration than wood decaying basidiomycetes (Daniel and Nilsson, 1998).

In comparison with soft-rot fungi, other wood decay fungi have a relatively narrow spectrum of growth conditions. The optimum moisture content for growth is between 35% and 50%. A high moisture content of wood inhibits fungal growth due to oxygen restriction in the wood. However, wood decay fungi have been frequently observed in historic buildings exposed to outdoor conditions (Palanti et al., 2012; Pilt et al., 2009; Sterflinger, 2010). Brown- and white-rot fungi were observed in the historic buildings in Chilean wooden churches designated as a World Heritage Site by UNESCO (Ortiz et al., 2014). Kisternaya and Kozlov (2012) found that the most common destructors of historic timbers in northern Russia were brown-rot fungi, such as *Serpula lacrimans, Coniophora puteana, Coriolellus sinuosus, Fibuloporia vaillanti,* and *Paxillus paunmoides*. The dry-rot fungus *Serpular* spp. occurred frequently in poorly ventilated spaces such as on the backside of floorboards or ceilings in basement wood with moisture content of 20–30%. Limited period of exposure to wetting conditions in a certain season did not prevent the invasion of historic buildings by wood decay fungi. It should also be mentioned that in extreme cases, some wood decay fungal mycelium grew on wood with 17.4% moisture content. Stienen et al. (2014) observed that *Antrodia xantha*, *Coniophora* spp., *Donkioporia expansa*, and *Gloeophyllum abietinum* are able to degrade wood samples below fiber saturation over the temperature range of 10–25 °C, suggesting that dry-type WCH are not always immune to fungal attack.

Deterioration of Underwater WCH by Biotic Agents

WCH submerged in seawater, lake, or mud for long periods can undergo structural and chemical alterations. Most water-saturated WCH with moisture content over 100–800%, are referred to as waterlogged wood. Some WAW were



(a) (b) (c)

FIGURE 12.11 (a) Excavation of underwater ancient wooden ship (AD 11–14C) in Korea, (b) soft-rot cavities in the waterlogged archeological woods, excavated from Sweden, (c) erosion bacteria in the waterlogged archaeological woods, excavated Germany. *(Unpublished files from Kim, Y.S.)*

found to be heavily degraded within only a few 100 years, while others have survived thousands of years of exposure and in some cases appear to be relatively well preserved (Bjordal et al., 2012; Kim and Singh, 2000; Clausen, 1996; Qiu et al., 2013; Schmitt et al., 2005). This suggests that age alone does not determine the extent of deterioration in the WCH. Most wet-type WCH were found in a relatively good condition even after prolonged periods of exposure to water-saturated environments. This is because the activity of faster wood-degrading fungi (Basidiomycete) is restricted due to wood being water-saturated. Instead, bacteria and soft-rot fungi played an important role in the degradation process of such woods (Fig. 12.11).

Soft-rot fungi are tolerant to a wide range of temperature, humidity, and pH conditions and can attack a variety of wood substrates (Daniel and Nilsson, 1998). Bacteria are very tolerant to conditions considered extreme for most wood-decay fungi, for example high lignin and extractives content in wood, high preservative loading, and low levels of oxygen. Bacterial erosion has been found to be widespread in sediment where the archaeological woods had been buried (Fig. 12.11). Degradation of WAW by erosion bacteria is followed by scavenging bacteria in some situations (Fig. 12.12) (Kim and

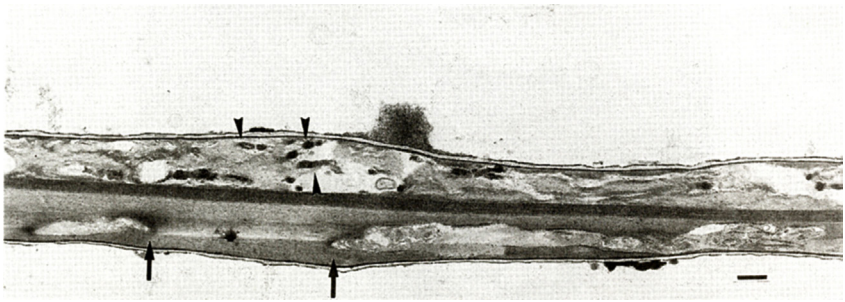


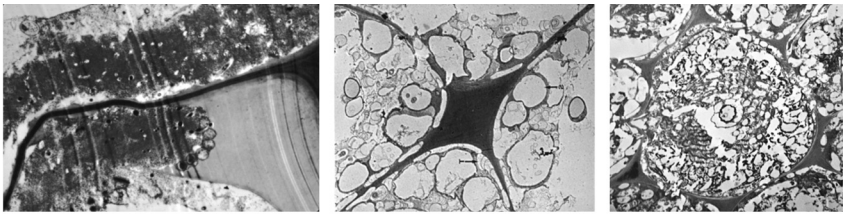
FIGURE 12.12 Degradation of S2 layer by erosion bacteria (arrows) in *C. camphora* wood in an excavated wooden ship in Korea. Scavenging bacteria (arrowheads) are present within the degraded wall material (bar = 1 μm) (Kim et al., 1996).

Singh, 1994), some being sulfur-reducing bacteria, which produce hydrogen sulfide *in situ* (Fors et al., 2012). Co-occurrence of bacterial erosion, bacterial tunneling, and soft rot has also been noted in buried and waterlogged woods. Detailed information on the micromorphology of bacterial degradation can be found in a previous chapter in this book (Singh et al., 2014).

It is now widely accepted that attack by erosion bacteria is the main cause of destruction of WAW, as these bacteria are capable of degrading wood under conditions of limited oxygen availability (Huisman et al., 2008; Singh, 2012). Attack of WAW by erosion bacteria is mainly confined to the secondary cell wall (Nilsson et al., 2008), whereas attack of WAW walls by tunneling bacteria was found to extend to the middle lamella. This suggests that tunneling bacteria can degrade all areas of lignified cell walls, including the highly lignified middle lamella (Singh et al., 2003). Considering that tunneling bacteria have a relatively high requirement for oxygen (Daniel et al., 1989), the attack on WAW by tunneling bacteria is likely to have occurred before the objects were completely submerged in the ocean or mud. The micromorphology of waterlogged archaeological bamboo was similar to that for WAW. However, typical soft-rot cavities, which were frequently observed in waterlogged archaeological woods, were not found (Cha et al., 2014).

The degradation of WAW walls did not occur uniformly; relatively-sound tracheids and fibers were found among extensively degraded regions. Degradation occurred mainly in tracheids and fibers, whereas vessel walls were relatively intact. Separation of the secondary wall layers from middle lamella was frequently observed in all cell types. Degradation occurred preferentially in the S2 layer, whereas middle lamellae were mostly intact. The degraded tracheids and fibers showing a granular appearance were distinctly porous and lacked birefringence when observed under a polarized microscope, indicating a loss in the crystalline cellulose. In hardwoods, vessels in WAW were more intact compared to other cell types, such as fibers and parenchyma cells. In advanced stages of degradation, vessel walls were also degraded with only the middle lamella remaining. Ultrastructural examination on the WAW showed that cell outlines were distinguishable despite extensive losses in the integrity of the bulk of cell wall material. This is primarily because middle lamellae remained usually intact, although often highly convoluted. Resistance of middle lamella to erosion bacteria under waterlogged situations is likely to be due to its high lignin concentration. Compound middle lamella and vessel walls in woods are reported to exhibit higher lignin concentration compared to fiber secondary walls (Kim et al., 2008) and the composition of lignin is also different from the secondary cell walls (Faix, 1991; Fengel and Wegener, 1984).

Microscopic examinations showed that bacterial attack was the main cause of degradation of waterlogged woods and bamboos (Fig. 12.13) (Kim, 1989; Kim et al., 1996; Kim and Singh, 2000; Blanchette, 2000; Singh, 2012; Cha et al., 2014). Cell wall components, particularly lignin, were the main factors that influenced microbial degradation in underwater WCH. However, other



(a) (b) (c)

FIGURE 12.13 Degradation of waterlogged archeological woods and bamboos. *((a) Degradation of tracheids by erosion bacteria, (b) co-occurrence of tunneling bacteria, (c) soft-rot fungi and degradation of waterlogged archaeological bamboos by erosion bacteria.)*

factors, such as leaching of degraded cell wall materials and impregnation of cell walls by inorganic substances over time, may also affect the degree of deterioration of WAW. Nonbiological factors, such as salinity, temperature, nutrients, composition of sediment layers, and other factors, can also influence the rate of microbiological decay of WAW (Bjordal et al., 2012). Further studies are needed to determine whether abiotic reactions in seawater, such as hydrolysis, might also partly contribute to degradation of wood components of WAW, although it is now well documented that microbiological factors are the primary cause for the deterioration of woods subjected to waterlogging.

SOME REMARKS ON THE CONSERVATION OF WCH

Conservation includes maintenance, preservation, restoration, reconstruction, and adaptation; and commonly a combination of these, whereas preservation means maintaining the fabric of a place in its existing state and retarding deterioration. Conservation means all the processes of looking after a place so as to retain its cultural significance (Jokilehto, 2005). Consequently, conservation work is a very time-consuming, labor-intensive, and fairly expensive process. The conservation of WCH has been on empirical approaches rather than hard scientific data because each WCH is unique in terms of wood species, environment, deterioration rate, and manufacturing techniques. Individual assessment, evaluation, and solutions are required for the conservation and restoration of WCH. The application of general standard solutions into the conservation of WCH should be carefully exercised. In practice, different people, each with different sets of skills and paradigms, have carried out the selection and application of conservation treatments. The conservation professionals strive to select methods and materials that do not adversely affect cultural property. Techniques of conservation of WCH range from artistic forms to scientific technologies. Holistic approaches are required for WCH conservation; combination of curator's focus on context, archaeologist's deep awareness of the past, and wood scientist's comparative methods are needed for a rational integration of technical and scientific knowledge for effective conservation (Peterson, 1990). The mastery

of an artist as well as the analytical abilities of a scientist is required. To avoid miscommunication and misunderstanding among the conservators, curators, the governmental authorities, and wood scientist, a coordinated approach for conservation should be developed. For the conservation of WCH, some recent books can be referred, for example, Bjordal et al. (2012), Christensen et al. (2012), Lucejko (2010), Unger and Schniewind (2001), and the proceedings from the International Council of Museums.

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